

There is More to Attention: Statistical Filtering Enhances Explanations in Vision Transformers

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Filter before you aggregate.

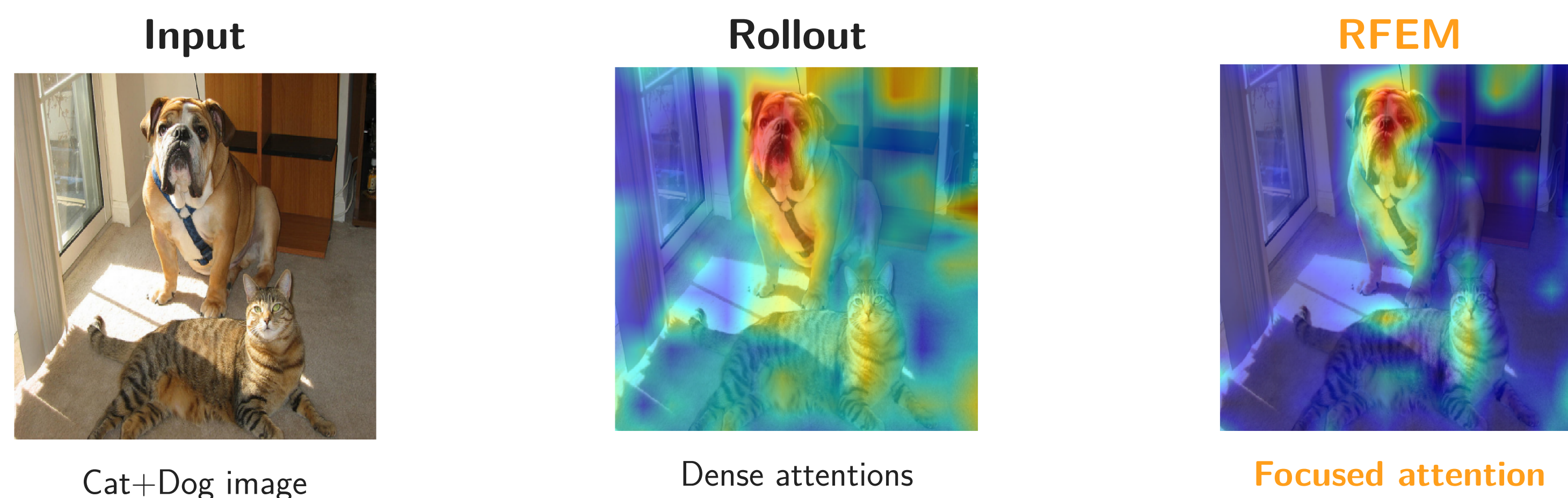
RFEM filters Vision Transformer attentions, producing explanations that are simultaneously faithful and more human-aligned.

The Core Idea

ViT attentions are dense across all layers and heads.

Standard Rollout → Aggregates Everything → Noisy Explanations

Our Proposal → Statistical Filtering → Clean Explanations



Why RFEM?

Dense attention matrices contain many weak token interactions that accumulate during rollout, producing diffuse explanations.

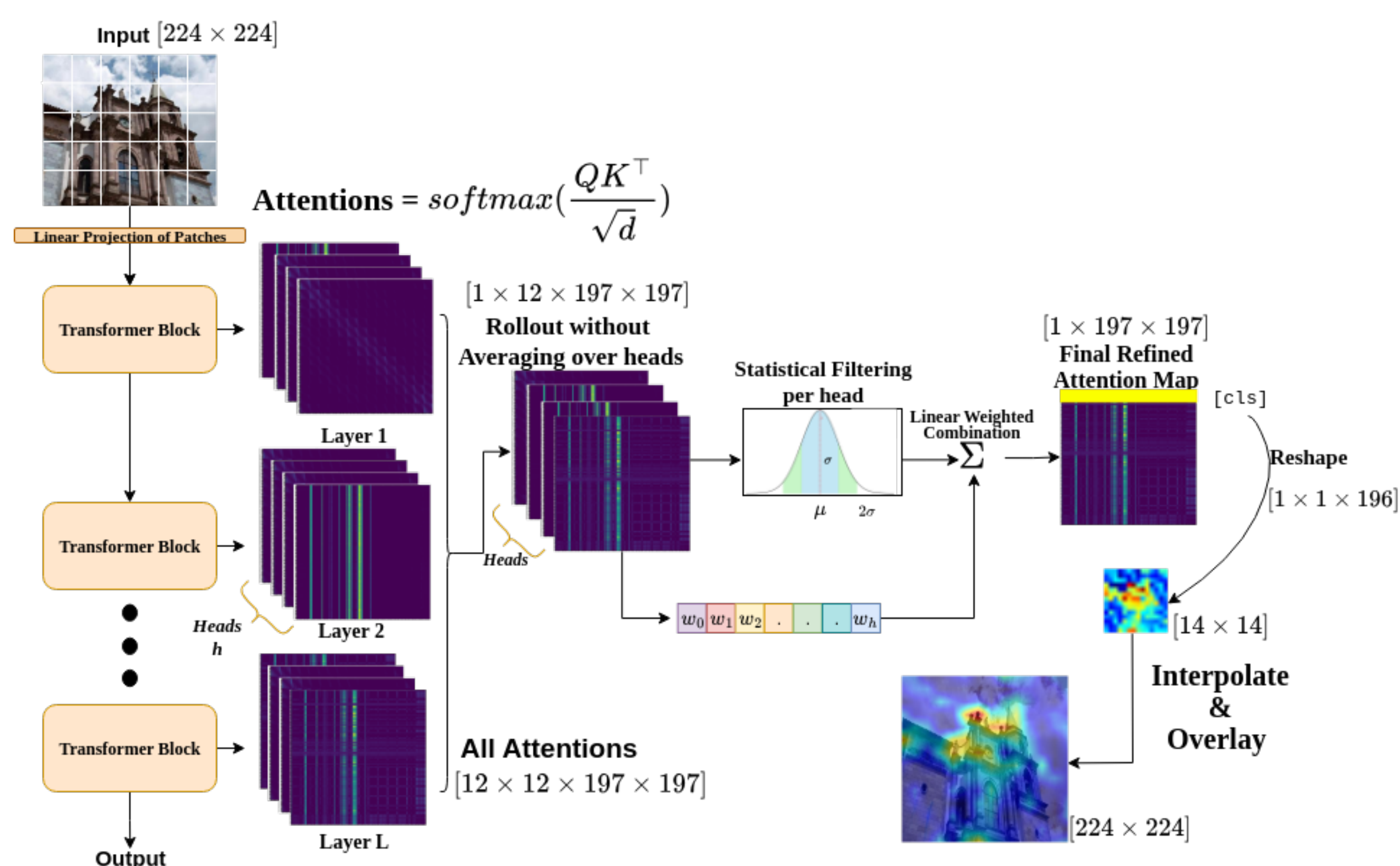
✓ K - σ thresholding to remove diffuse background attentions

✓ Head-wise adaptive filtering preserves only informative token interactions

✓ Better localization and alignment with human gaze

✓ 3x to 7.5x faster inference than complex attribution methods (e.g., AttnLRP) in a single pass

The RFEM Framework



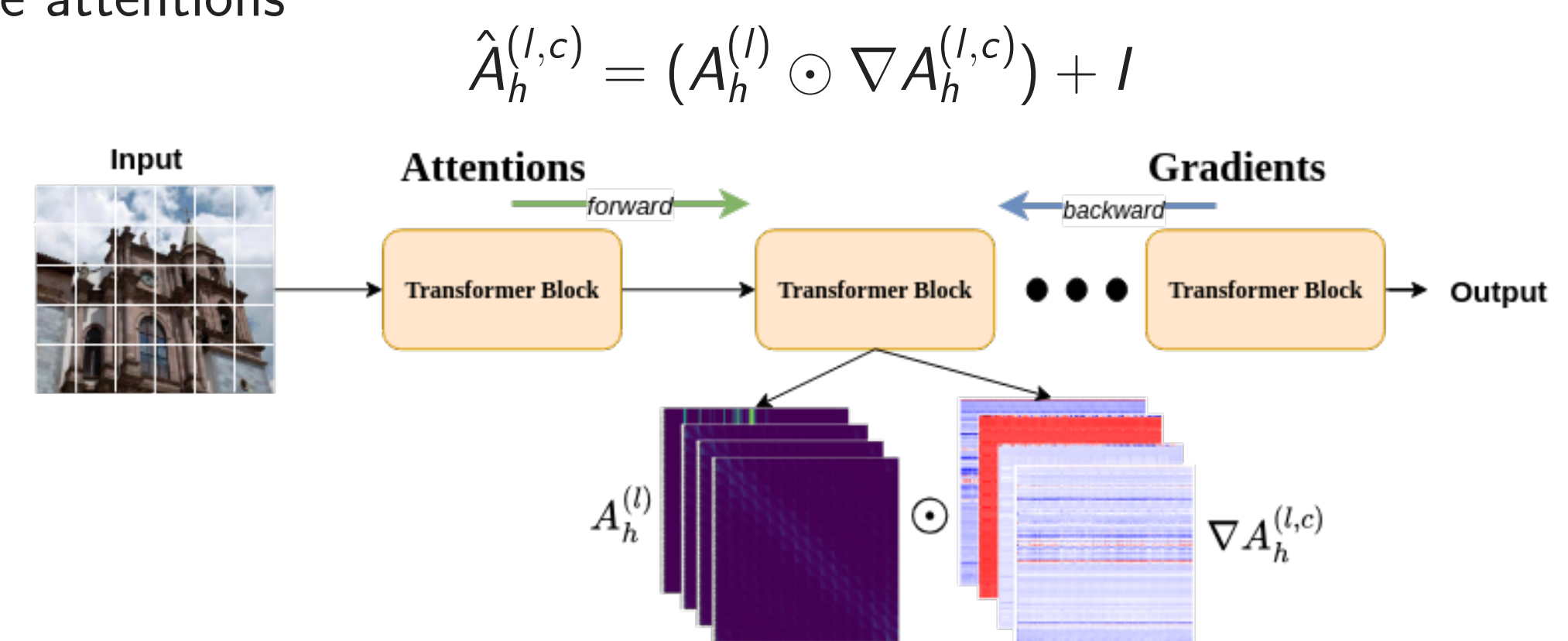
For attentions from every layer l and head h add identity matrix I for skip connections → sequential multiplication across all L :

$$A_h^{(l)} = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right) \quad \hat{A}_h^{(l)} = A_h^{(l)} + I \quad \hat{A}_h = \hat{A}_h^{(L)} \cdot \hat{A}_h^{(L-1)} \dots \hat{A}_h^{(1)}$$

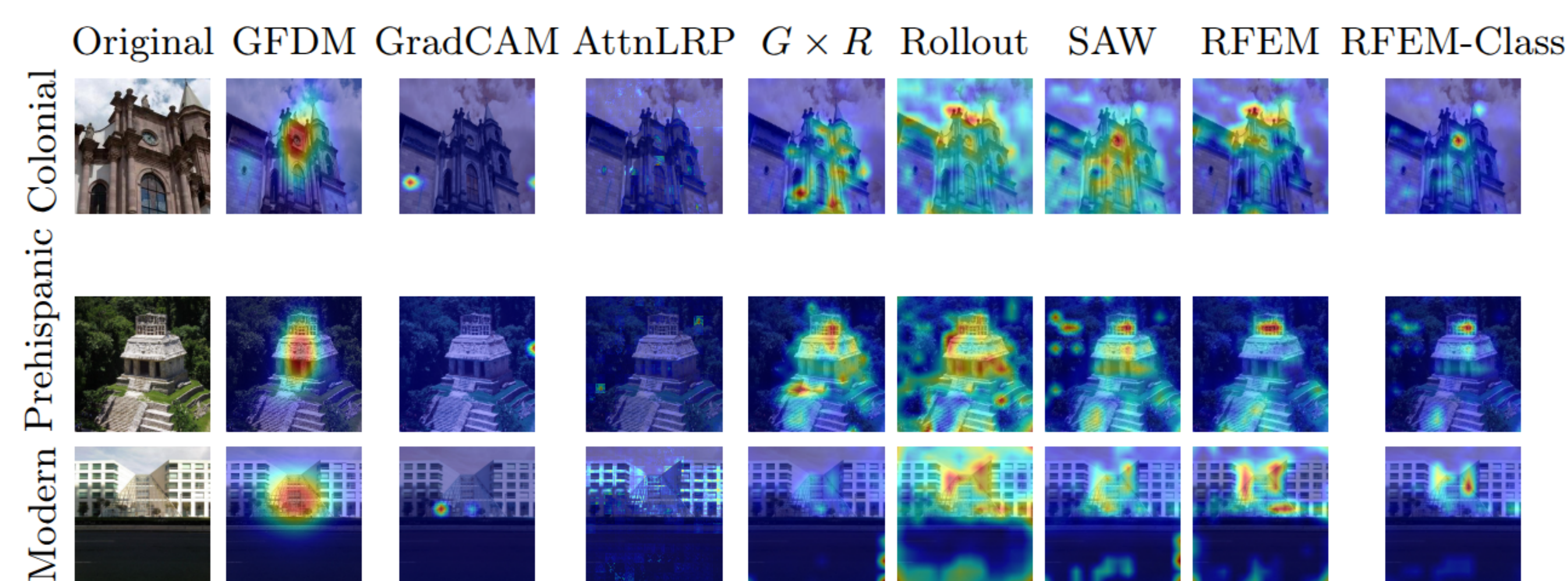
Apply K - σ thresholding and then fuse the heads using weighted average:

$$\bar{A}_h = \begin{cases} 1, & \text{if } \hat{A}_h(i, j) \geq \mu_h + K\sigma_h \\ 0, & \text{otherwise} \end{cases} \quad A_{\text{rfem}} = \sum_{h=1}^H w_h \cdot \bar{A}_h \quad \text{where } w_h = \max(\hat{A}_h)$$

RFEM-Class: To isolate specific objects we use the gradients $\nabla A_h^{(l,c)}$ of class c to modulate the attentions

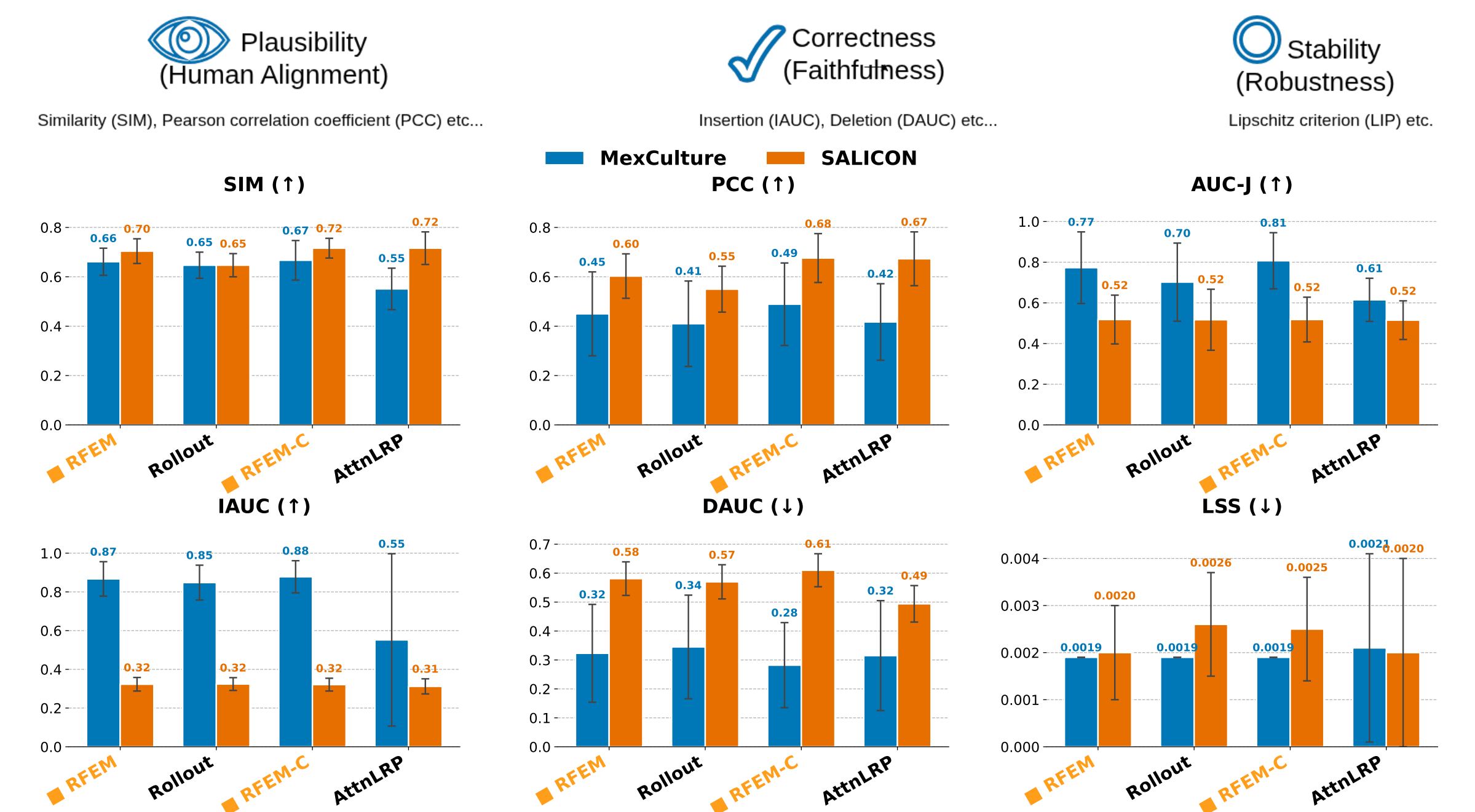


Alignment with Human Gaze



Qualitative comparison showcasing RFEM closely aligning with human Gaze Fixation Density Maps (GFDMs).

Quantitative Comparison



Limitations

- ✓ Threshold K is dataset dependent
- ✓ Class-specific mode requires gradients
- ✓ Evaluated primarily on ViTs

References

- [1] Fuad, K.A.A., Martin, P.E., et.al. Features understanding in 3d CNNs for actions recognition in video. IPTA 2020
- [2] Abnar, S. and Zuidema, W., Quantifying attention flow in transformers. ACL 2020
- [3] Achibat, R., Hatefi, S.M.V., et.al. AttnLRP: Attention-Aware Layer-Wise Relevance Propagation for Transformers. ICML 2024

